Same Models, Different Conditions

Why Quantitative Methods Need Not Deliver Quantitative Results for a Retail Trader: Sixteen Pre-Registered Statistical-Arbitrage Tests on Regulated Crypto Perpetuals

Guillermo Hernández Vidal*

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Abstract

Why does using the methods of a quantitative fund not necessarily let a small investor reproduce its results? We study the question with sixteen statistical-arbitrage tests on Kraken's crypto perpetuals, a MiFID II-regulated venue (454 contracts, including 290 delisted; hourly data, 2022–2026): twelve cointegration-based pairs variants, two follow-ups, an Avellaneda–Lee residual strategy and a funding-carry portfolio, each with a pass rule fixed before execution. None passed. We keep apart observed return, the premium attributable to each mechanism and statistical evidence of an edge. Avellaneda–Lee earned 0.44 gross per trade against 0.37 of costs. Carry returned +19.2% in the twelve months with real funding, mostly from price moves; funding minus costs was about 2.8% a year on capital, and the strategy was rejected by a 10.07% drawdown in the period with estimated funding. Four quantitative arguments—costs per execution, effective breadth, data sufficiency and premium versus risk—show how implementation constraints can absorb a theoretical edge or keep it statistically undetectable; some depend on capital, others on the market, its data or our design. The conclusions are limited to these strategies, this venue, retail costs and 2022–2026 data, and failing a test does not show that the true return is zero.

Keywords: statistical arbitrage; cointegration; funding carry; perpetual futures; deflated Sharpe ratio; pre-registration; retail investors; MiFID II.
JEL: G11, G12, G14, G15.

1 Introduction

Statistical arbitrage—trading the temporary divergence of prices that are expected to move together—is one of the oldest quantitative strategies. Pairs trading earned excess returns in U.S. equities for decades (Gatev et al., 2006; Vidyamurthy, 2004), although its profitability has declined and is sensitive to trading costs (Do and Faff, 2010, 2012; Krauss, 2017). Avellaneda and Lee (2010) generalised the idea to residuals with respect to common factors and reported net Sharpe ratios of about 1.44 on a broad universe of U.S. equities over 1997–2007, falling to about 0.9 in 2003–2007. Crypto perpetual futures look like a natural new field: they trade around the clock, offer hundreds of contracts, and pay a funding rate that has been shown to carry a large and volatile premium (Schmeling et al., 2023; Christin et al., 2022); carry strategies in general earn a premium across asset classes (Koijen et al., 2018), and crypto markets have shown large, persistent price gaps between venues (Makarov and Schoar, 2020).

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This paper asks a narrower question, from the point of view of an individual in the European Economic Area (EEA): *why does using the methods of a quantitative fund not necessarily allow a small investor to reproduce its results?* The question is not whether individuals can make money trading—nothing here says they cannot—but whether the particular advantage that statistical arbitrage offers a fund survives the conditions under which one person implements it: retail capital, retail costs, a regulated venue and the data that one person can collect. We built a complete, local and open trading-research system for Kraken's linear perpetuals and evaluated 16 tests whose hypotheses, parameters and pass/fail rules were committed to a version-control repository *before* each test was run (Nosek et al., 2018). All 16 failed the pre-registered criterion.

A negative result is only useful if it explains *why*. Our contribution is to organise the evidence into four quantitative arguments, each stated as an equation with every symbol defined and filled in with our own numbers, and each contrasted with what an institutional participant would face:

(a) a *cost inequality*: the expected gross gain per trade must exceed the cost of the executions it requires;

(b) *breadth*: by the fundamental law of active management, the information ratio grows with the square root of the number of independent bets, and our tradable universe is tiny;

(c) *statistical power*: with twelve months of data, a strategy needs a very high Sharpe ratio to be distinguishable from luck once multiple testing is accounted for;

(d) *premium versus risk* in funding carry: a net premium of a few percent a year is small relative to the price risk of a handful of legs, and capital and diversification change that ratio.

Throughout, we separate three kinds of constraint, because they have different remedies: those that depend on *capital* (the size of each position, how many legs fit within a leverage limit, the weight of minimum and fixed costs); those that depend on the *market and its data* (the number of liquid contracts, the common movement among them, the length of the funding history); and those that *we imposed* on our own implementation (taker execution at hourly prices, no spot positions, a 10% drawdown limit). More capital would relax the first group but not the second or the third, and institutions face costs, execution problems and capacity limits of their own. The paper can therefore be read at two levels. Section 2 states the question, the constraints and the scope of the conclusion in plain language, with our own numbers; the remaining sections give the regulatory setting, the data, the methodology, the results and the formal version of the four arguments, which is what allows a reader to evaluate that conclusion.

The scope of our conclusions is deliberately limited. We do not claim that statistical arbitrage is impossible in general, nor in crypto. We show that *these* strategies, on *this* venue, with *retail* costs and *2022–2026* data did not meet a demanding but standard evidential bar, and we quantify the mechanisms. Not meeting that bar is not evidence that the true return of a strategy is zero; it means that our data cannot distinguish it from zero with the confidence we required. Several limitations—only twelve months of real funding data, a small universe, a carry verdict decided at the margin in the estimated-funding period, and the sensitivity of one variant to small changes in estimated spreads—are discussed explicitly in Section 9.

Section 2 presents the problem without technical notation, Section 3 describes the regulatory setting, the choice of market and our design decisions, Section 4 the data, Section 5 the methodology, Section 6 the results, Section 7 the four arguments and their synthesis, Section 8 the discussion, Section 9 the limitations, Section 10 the pre-registered future work and Section 11 concludes.

2 Same models, different conditions: the retail investor's problem

This section can be read on its own; the formal arguments, with their equations and assumptions, are in Section 7.

What we tried to replicate. Statistical arbitrage, as quantitative funds practise it, does not bet on whether markets rise or fall. It looks for prices that usually move together, buys the one that has fallen behind, sells the one that has run ahead and closes both when they converge; or, in funding carry, it collects a payment that some contracts make to others. Each bet has a small expected profit, so the approach relies on three things a fund has in abundance: many bets at the same time, trading costs that are small relative to that profit, and long histories on which to tell a real advantage from luck. We applied the same models—cointegration tests, factor residuals, funding rankings—with an individual's resources: a simulated account of USD 1,000, positions of about USD 100 per leg, and the data one person can collect from a public interface.

Capital decides the size of each position, and size decides the weight of costs. With USD 1,000 and a gross exposure capped at twice the capital, three long and three short positions of USD 100 make a gross exposure of 60% of the capital, and ten of each would reach the cap; more legs would have to be smaller. On crypto perpetuals such small positions are possible: the minimum order size was the first reason to exclude a contract in only 17 of about 46,000 contract-weeks we examined, and fees are a percentage of the amount traded. Holding the same small long–short book in European shares would be different: at a widely used retail broker, an order costs at least EUR 1.25 in commissions (EUR 3.00 on its simplest tariff), so a EUR 100 order pays 1.25–3% of its value before any spread, against the 0.08% per execution that we model for perpetuals (Section 3). Charges that are negligible for a fund can decide whether such a book is practical for a small account.

A small expected profit can vanish when every leg is executed. An Avellaneda–Lee trade needs six executions: opening and closing the asset and its two hedges in bitcoin and ether. In our data the average trade earned USD 0.44 before costs for every USD 100 placed in the asset, and the six executions cost USD 0.37:

$$\text{net edge per trade} = \text{gross edge per trade} - \sum \text{cost of each execution.} \tag{1}$$

What remained, USD 0.07 per USD 100 before funding, is about 15% of the gross edge and tiny next to the swing of a single trade, whose standard deviation was about USD 19. The model may have captured something real; most of it was absorbed by implementation.

Many prices are not many opportunities. Our database holds about 7.6 million hourly prices of 454 contracts, but the abundance is misleading. In a typical week only about ten contracts were liquid enough to trade (9.9 on average), and they largely rose and fell together: even after removing their exposure to bitcoin and ether, the returns of the contracts in the carry portfolio kept an average correlation of 0.21. Out of 2,041 pairs tested over 221 weeks, 153 passed a 5% cointegration test in both directions, when about 102 would pass by chance alone; after controlling for false discoveries and checking stability, 13 pairs survived, in 12 of the 221 weeks. What matters is not the number of prices or of assets but the number of *independent* bets a strategy can make, and correlated assets provide fewer of them than their count suggests.

A profitable backtest is not evidence of an edge. Several of our tests made money: the best pairs variant earned +3.2% over four years, and funding carry +19.2% in its twelve months with real funding data. A backtest is one draw from a noisy process, and the more variants are tried, the more likely the best of them looks good by chance. The usual yardstick, the Sharpe ratio (average return divided by its volatility), turns into evidence only with time:

$$t \approx \mathrm{SR}_{\mathrm{annual}} \times \sqrt{\text{years of data}}, \tag{2}$$

which must reach about 1.65 for 95% confidence with a single test, and more when several variants have been tried. With twelve months, a strategy therefore needs an annual Sharpe ratio of about 1.65 just to be told apart from luck; carry reached 1.32. Falling short does not show that the true return is zero; it shows that our data cannot tell.

Three different numbers in the carry result. Funding carry shows why observed return, the premium of the mechanism under study and statistical evidence must be kept apart. The portfolio returned +19.2% in the real-funding period, but most of it came from favourable price moves of the legs. The mechanism we were studying—funding collected minus trading costs—contributed USD 27.5 on USD 1,000 of capital in 355 days, about 2.8% a year. Statistically, the deflated Sharpe ratio was 0.904 against the 0.95 required, so on the real data alone the result was *inconclusive* under our rule. The hypothesis was rejected because the simulation of the whole period, which needs estimated funding before October 2025, fell 10.07% against a 10% limit.

What depends on what. These obstacles do not share a single cause. Position size, how many legs fit within the leverage limit and the weight of minimum costs depend on *capital*. The small number of liquid contracts, their common movement and the short funding history depend on the *market and its data*; more capital would not change them. Taker execution at hourly prices, the absence of spot positions and the 10% drawdown limit are *our own design choices* (Section 3); other choices could do better or worse. Nor are institutions free of these problems: they pay costs, suffer execution slippage and cannot place large amounts in small contracts. Section 7 turns this intuition into four quantitative arguments—costs, the effective breadth of opportunities, the sufficiency of the data and the premium relative to risk—each with its equation, assumptions and evidence, and Section 7.5 shows how they combine.

3 Regulatory setting, choice of market and design decisions

This section separates three things that are easily confused: what the rules allow a retail client to do, what is economically practical with our capital, and what we decided ourselves. Figures on fees and access were checked on the dates indicated; they change, and the sources should be consulted again before relying on them.

3.1 Regulation and access

Two European regimes are relevant. The Markets in Crypto-Assets Regulation (MiCA) governs crypto-assets and the providers of crypto-asset services (European Parliament and Council, 2023); its provisions apply in full from 30 December 2024, with transitional periods for providers already operating under national law (European Securities and Markets Authority, 2024b). MiCA explicitly excludes from its scope crypto-assets that qualify as financial instruments under MiFID II (Article 2(4)), and ESMA has issued guidelines on that qualification (European Securities and Markets Authority, 2024a). Derivatives on crypto-assets, such as perpetual futures, are offered to EEA clients as financial instruments under the Markets in

Financial Instruments Directive (MiFID II; European Parliament and Council, 2014), which brings investor-protection duties (e.g. assessments of the client's knowledge before derivatives can be traded).

Access was not, in itself, the binding constraint. A retail client in the EEA can open an account with a broker, buy and sell shares and, in a margin account and with the broker's approval, sell them short; uncovered short sales of shares are restricted by the Short Selling Regulation, which requires a borrowing or locate arrangement (European Parliament and Council, 2012). For crypto perpetuals the constraint is narrower: they must be offered by a MiFID II investment firm, and an authorisation under MiCA alone is not enough. In a comparison of 21 platforms that we compiled on 23 September 2026 from an aggregator of MiCA registrations and fee schedules (PerpFinder, 2026) and from company announcements—secondary sources, not official registers—three could serve a retail client resident in Spain (and, through the EU passport, elsewhere in the EEA) with crypto perpetuals, enough contracts and an API for automation: Kraken (more than 290 perpetual contracts), OKX Europe (perpetual-style contracts with a five-year expiry and their own order book; between about 85 and 155 contracts, depending on the source) and Backpack EU (more than 40). According to the same sources, the other eighteen failed at least one of these requirements.

We use Kraken because it offers EEA clients linear (USD-margined) perpetual futures through a MiFID II-regulated entity, Payward Europe Digital Solutions (CY) Ltd, supervised by the Cyprus Securities and Exchange Commission; the offering was announced in May 2025 (Kraken, 2025). For a researcher, the venue has three further advantages: a public API that still serves the price history of delisted contracts (which allows survivorship-free backtests), hourly funding settlement, and a published fee structure. Throughout, we assume the base-tier taker fee of 0.05% per execution (Kraken, 2020); Binance publishes the same base-tier figures for its USD^{S}-M perpetuals (0.02% maker, 0.05% taker), with lower fees for higher-volume tiers (Binance, 2026). Kraken's current fee schedule, consulted on 21 October 2026, confirms the base tier—0.02% maker and 0.05% taker below USD 5 million of futures volume in 30 days—and falls to 0.0125% taker and a maker *rebate* of 0.006% above USD 5 billion (Kraken, 2026); the page does not state whether exactly this table applies to the European entity, which our earlier comparison suggests but we have not confirmed in an official document. Kraken is not presented here as the best venue: among the three authorised options in our comparison it was the one with the largest number of perpetual contracts, a global order book shared with non-European clients and hourly funding. OKX Europe and Backpack EU had the same base fees in that comparison but fewer contracts; we did not test them.

3.2 Economic constraints

What decides whether a long–short book is practical with USD 1,000 is the cost of each order relative to its size. On Kraken perpetuals fees are proportional to the amount traded, short positions need no borrowing, and one collateral balance serves every leg; the minimum order size was the first reason to exclude a contract in only 17 of about 46,000 contract-weeks of our weekly universe selection. Perpetuals add a payment that shares do not have, funding, which can be a cost or an income and which the backtests include hour by hour.

For European shares we did not run a backtest, but the fee schedule of a widely used broker shows the order of magnitude. At Interactive Brokers Ireland, consulted on 21 October 2026, an order on German exchanges costs 0.05% of its value with a minimum of EUR 1.25 on the tiered tariff (plus exchange and clearing fees), and EUR 3.00 on the fixed tariff, which is also the minimum listed for Spanish shares (Interactive Brokers Ireland, 2026a). With EUR 100 per leg, the minimum alone is 1.25–3% of the order, about 16–38 times the 0.08% per execution that we model for perpetuals, and an Avellaneda–Lee trade with six executions would pay at least EUR 7.50 in commissions against the gross edge of USD 0.44 per USD 100 that the

strategy earned on perpetuals. The minimum stops dominating only for orders of EUR 2,500 (tiered) to EUR 6,000 (fixed), so a book of three legs per side at that size needs EUR 15,000–36,000 of gross exposure. Short positions add a borrowing fee that depends on each stock ("general collateral" or "hard to borrow") and, at the same broker, euro balances from short sales earn no interest below EUR 90,000 (Interactive Brokers Ireland, 2026b). We have *not* estimated borrowing fees, exchange and clearing fees, market-data subscriptions or the cost of survivorship-free historical data for shares; these remain pending, and the comparison above is therefore a lower bound on the cost of the equity route. None of this says that an individual cannot trade shares; it says that a many-legged long–short book of small positions is expensive to hold there.

3.3 Decisions we imposed on the system

Several features of our results come from our own design, not from regulation or from the market, and they should not be read as evidence about what is possible: a single venue (Kraken); only linear crypto perpetuals; no spot positions and no transfers between accounts, so the classic "cash-and-carry" (long spot, short perpetual) is outside our scope—which does not show that cash-and-carry is reserved for institutions; execution only as a taker at hourly closing prices, at the base fee tier; a simulated capital of USD 1,000 with legs of 10% of capital (three per side in carry); gross leverage of at most 2, net exposure of at most 10% of capital, a stop-loss of 1.5% of capital per position (per leg in carry) and an emergency stop at a 10% drawdown; and the universe filters of Section 4. Other choices—passive execution, a second venue, spot legs, a looser drawdown limit—could change the results in either direction; apart from the registered variants, we did not test them. The system runs only in simulation mode. Nothing in this section is legal advice.

4 Data

Source and coverage. All data come from Kraken's public futures API (no keys): instrument metadata, hourly candles of three kinds (trades, mark price and index price), order-book snapshots and the funding-rate history. The sample contains **454 crypto linear perpetuals**, of which 164 were listed and 290 had been delisted at the time of download, from the first listings on 23 March 2022 to 17 October 2026. Non-crypto contracts (equities, commodities, currencies, indices and stablecoins) are excluded using the API's `category` field.

Survivorship bias. A backtest that uses only the contracts alive today overstates performance, because the contracts that died along the way disappear from the sample. We keep every delisted contract, cut its series two days after its last trade (afterwards the API returns flat candles), treat renamed assets (e.g. MATIC/POL) as the same asset so that they are never paired, and close any open position at the last available price if a contract stops trading.

Cleaning and the weekly universe. The tradable price is the last trade of each hour, replaced by the mark price when the hour had no trades or the trade deviated by more than 2% from the mark price. Each Monday at 00:00 UTC (a *cut-off*) we rebuild the universe using only data before the cut-off: median 30-day daily volume of at least US$1 million, mean quoted spread of at most 5 basis points, sufficient order-book depth, at least 120 days since listing, at most 1% of hours without any price, at most 5% of hours without trades, and (for the pairs strategies only) an absolute funding rate of at most 30% a year. Spreads come from order-book snapshots where available and, otherwise, from a log–log regression of spread on volume; because the order-book history is short, the filter applied to past weeks uses spreads measured in October 2026, a bias we document. There are 221 weekly cut-offs, from 25 July 2022 to 12

October 2026; the universe of the pairs strategies contained on average **9.9 contracts** (between 0 and 19).

Funding: real and estimated. Funding is the periodic payment between longs and shorts that keeps the perpetual close to its index price (He et al., 2022; Ackerer et al., 2026). On Kraken it is paid every hour; let $r_{i,h}$ denote the relative rate of contract i in hour h (a fraction of notional; positive values mean that longs pay shorts). The API only returns about twelve months of history, so real funding is available from **15 October 2025**. Before that, we estimate it from Kraken's formula, as the average premium of the mark price over the index price divided by 8, capped at $\pm 0.5\%$ per hour, and scale it by a factor k:

$$\hat{r}_{i,h} = k\,\tilde{r}_{i,h}, \qquad k = \frac{\sum_{i,h} r_{i,h}\,\tilde{r}_{i,h}}{\sum_{i,h} \tilde{r}_{i,h}^{2}}, \tag{3}$$

where $\tilde{r}_{i,h}$ is the unscaled estimate and the sums run over all hours with real funding of the 25 contracts with at least US$1 million of median daily volume (220,323 hours). The least-squares factor is $k = 0.42$. Fitting on the first six months and testing on the next six gives an out-of-sample bias of -1.8% a year, a mean absolute error of 11.4% a year and an hourly correlation of 0.82. The estimate orders contracts well—the mean weekly Spearman correlation between the ranking by estimated and by real funding is 0.956—but its level is noisy, so every test reports the real-funding period separately.

5 Methodology

5.1 Walk-forward protocol and cost model

At each cut-off the strategy re-estimates everything with data before the cut-off and trades during the following week (a walk-forward design; on the backtest-overfitting risks it mitigates but does not remove, see López de Prado, 2018); signals are computed at the close of each hour and executed at that close. The cost of one execution (opening or closing one leg) of contract i, as a fraction of the traded notional, is

$$c_i = f + \max(h_i,\, h_{\min}) + s, \tag{4}$$

where $f = 0.05\%$ is the taker fee, h_i is half the mean quoted spread of contract i, $h_{\min} = 0.02\%$ is a floor and $s = 0.01\%$ is slippage; for almost every contract $c_i = 0.08\%$. Funding is paid hourly: a position of signed quantity q (positive for a long) in a contract with price P_h receives $-\operatorname{sign}(q)\,|q| P_h\, r_{i,h}$ in hour h. Risk limits are those of the author's risk file and were never changed: a position is closed if it loses more than 1.5% of capital; no new positions are opened after a daily loss of 2%; a drawdown of more than 10% from the peak stops the system (emergency stop); gross leverage is at most 2 and net exposure at most 10% of capital.

5.2 Pairs trading by cointegration

Let $y_t = \ln P_t^{y}$ and $x_t = \ln P_t^{x}$ be the log prices of two contracts in hour t of a formation window of $W = 90$ days. *Pre-selection.* Hourly returns of the universe are standardised and summarised by principal component analysis (PCA; Jolliffe, 2002) with $K = 5$ components; each asset is represented by its vector of loadings, and OPTICS (Ankerst et al., 1999) clusters assets with similar loadings (minimum cluster size 2). Only pairs within a cluster are tested.

Cointegration (Engle and Granger, 1987). We estimate by ordinary least squares

$$y_t = \alpha + \beta\, x_t + u_t, \tag{5}$$

where α is a constant, β the hedge ratio and u_t the spread, and test whether u_t has a unit root with the augmented Dickey–Fuller regression (Dickey and Fuller, 1979)

$$\Delta u_t = \gamma\, u_{t-1} + \sum_{j=1}^{p} \phi_j\, \Delta u_{t-j} + \varepsilon_t, \tag{6}$$

where $\Delta u_t = u_t - u_{t-1}$, the null hypothesis $\gamma = 0$ means no cointegration, the lag order p is chosen by the Akaike criterion and ε_t is white noise; p-values use MacKinnon's critical values for estimated residuals (MacKinnon, 2010). Because the result depends on which series is on the left, we test both directions and take the larger p-value, $p_{\text{pair}} = \max(p_{y|x}, p_{x|y})$.

Multiple testing. With m pairs tested in a week, we control the false discovery rate with the Benjamini–Hochberg procedure (Benjamini and Hochberg, 1995): sorting $p_{(1)} \le \cdots \le p_{(m)}$, we declare cointegrated the k^* pairs with smallest p-values, where $k^* = \max\{k : p_{(k)} \le (k/m)\,\bar{q}\}$ and $\bar{q} = 5\%$.

Spread filters. With s_t the spread of the selected regression: the Hurst exponent H, estimated as the slope of $\ln \sigma(\tau)$ on $\ln \tau$ with $\sigma(\tau)$ the standard deviation of $s_{t+\tau} - s_t$ for $\tau = 2, \ldots, 100$ hours (after Hurst, 1951), must satisfy $H < 0.5$ (mean reversion); the half-life, obtained from the AR(1) fit $s_t = c + \varphi\, s_{t-1} + e_t$ as

$$\text{HL} = -\frac{\ln 2}{\ln \varphi}, \tag{7}$$

where φ is the autoregressive coefficient, must lie between 4 and 72 hours; the spread must cross its mean at least 20 times and its standard deviation σ_{eq} must be at least 0.5%; the pair must be cointegrated at 10% in each half of the window; $\beta > 0$; at most 10 pairs, and no asset in more than two.

Kalman filter. A Kalman filter (Kalman, 1960) with state $\theta_t = (\beta_t, \alpha_t)^\top$ following a random walk, $\theta_t = \theta_{t-1} + w_t$ with $w_t \sim N(0,\, \frac{\delta}{1-\delta} I)$, and observation $y_t = F_t \theta_t + e_t$ with $F_t = (x_t, 1)$ and $e_t \sim N(0, R)$, monitors breakdowns. With the one-step prediction error $v_t = y_t - F_t \hat{\theta}_{t|t-1}$ and its variance $f_t = F_t P_{t|t-1} F_t^\top + R$ ($P_{t|t-1}$ is the predicted state covariance), the parameters δ and R are chosen by maximum likelihood over a grid,

$$\ln L(\delta, R) = -\frac{1}{2} \sum_t \left[\ln(2\pi f_t) + \frac{v_t^2}{f_t} \right], \tag{8}$$

with $\delta \in \{10^{-10}, \ldots, 10^{-4}\}$ and $R \in \{0.01, \ldots, 1\} \times \sigma_{eq}^2$ (Harvey, 1989). The Kalman z-score is $z_t^K = v_t / \sqrt{f_t}$.

Signals. The equilibrium z-score is $z_t = (y_t - \beta x_t - \alpha)/\sigma_{eq}$. A position opens when $|z_t| \ge 2$ (selling the spread if $z_t > 0$), closes when $|z_t| \le 0.5$, and is stopped when $|z_t| \ge 4$, after three half-lives, or on a breakdown, $|z_t^K| \ge 4$; a pair stopped this way is not reopened until the next cut-off. An entry is skipped if the expected funding cost until exit exceeds 25% of the expected gain. Each pair uses 20% of capital (gross), split as $G/(1+\beta)$ in y and $\beta G/(1+\beta)$ in x, with G the gross notional.

5.3 Residuals with respect to factors (Avellaneda–Lee)

For each contract i and hourly log return $R_{i,t}$ in a window of $W = 60$ days we estimate

$$R_{i,t} = \beta_{i,0} + \beta_{i,B}\, R_{B,t} + \beta_{i,E}\, R_{E,t} + \epsilon_{i,t}, \tag{9}$$

where $R_{B,t}$ and $R_{E,t}$ are the returns of the BTC and ETH perpetuals (the factors), $\beta_{i,0}$ is a drift and $\epsilon_{i,t}$ the residual (Avellaneda and Lee, 2010). The cumulative residual $X_{i,t} = \sum_{u \le t} \epsilon_{i,u}$ is modelled as an Ornstein–Uhlenbeck process (Uhlenbeck and Ornstein, 1930), estimated through

its discrete version $X_{t+1} = a + b\,X_t + \zeta_t$, which gives the mean-reversion speed $\kappa = -\ln b$, the mean level $m = a/(1-b)$, the equilibrium standard deviation $\sigma_{eq} = \sqrt{\mathrm{Var}(\zeta)/(1-b^2)}$ and the half-life $\ln 2/\kappa$. The signal is the s-score $s_{i,t} = (X_{i,t} - m_i)/\sigma_{eq,i}$: buy the contract and sell its factor hedge when $s < -1.25$, sell it when $s > 1.25$, close a long when $s > -0.5$ and a short when $s < 0.75$, and trade only contracts with a half-life below $W/2$. Each asset leg uses 10% of capital; hedges of all positions are netted in one BTC and one ETH position, and net exposure is measured in beta.

5.4 Cross-sectional funding carry

Let $H = 168$ hours. At cut-off c, the signal of contract i is its mean funding over the previous week and the *next-week* funding is the analogous mean over the following week:

$$S_{i,c} = \frac{1}{H} \sum_{h \in [c-H,\, c)} r_{i,h}, \qquad N_{i,c} = \frac{1}{H} \sum_{h \in [c,\, c+H)} r_{i,h}. \tag{10}$$

The portfolio is short the three contracts with the highest $S_{i,c}$ and long the three with the lowest, each leg with 10% of capital (USD 100 per USD 1,000); with a *band*, a short is kept while it stays in the upper half of the ranking and a long while it stays in the lower half. The book is dollar-neutral by construction and is hedged in BTC and ETH only if its beta exposure exceeds 10% of capital; the 1.5% stop applies to each leg. The universe is the one above *without* the funding filter.

Persistence (step 1). Before any backtest we measured, for each cut-off, the Spearman correlation (Spearman, 1904) ρ_c between $S_{\cdot,c}$ and $N_{\cdot,c}$ across contracts, and the next-week spread

$$D_c = 8760 \left(\frac{1}{3} \sum_{i \in \mathcal{T}_c} N_{i,c} - \frac{1}{3} \sum_{i \in \mathcal{B}_c} N_{i,c} \right), \tag{11}$$

where $\mathcal{T}_c$ and $\mathcal{B}_c$ are the three contracts with the highest and lowest signal and 8760 annualises an hourly rate. The mean of ρ_c over T cut-offs, $\bar{\rho}$, is tested with a Newey–West standard error (Newey and West, 1987),

$$\widehat{\mathrm{Var}}(\bar{\rho}) = \frac{1}{T} \left[\hat{\gamma}_0 + 2 \sum_{l=1}^{L} \Big(1 - \frac{l}{L+1}\Big) \hat{\gamma}_l \right], \qquad L = \left\lfloor 4\,(T/100)^{2/9} \right\rfloor, \tag{12}$$

where $\hat{\gamma}_l$ is the sample autocovariance of ρ_c at lag l and L follows Newey and West (1994). The rotation τ is the weekly fraction of legs that must be replaced; from the tercile transition matrix, $\tau = 1 - (P_{HH} + P_{LL})/2$, with P_{HH} (P_{LL}) the probability that a contract in the top (bottom) tercile by signal stays there the next week. The registered rule required $\bar{\rho} > 0$ (two-sided 5%) and a mean D_c above the break-even spread derived in Section 7.1.

5.5 Evaluation, deflated Sharpe ratio and pre-registration

Let r_d be the daily returns of the capital curve. The Sharpe ratio is $\mathrm{SR} = \bar{r}_d/\hat{\sigma}(r_d)$ per day and $\mathrm{SR}\sqrt{365}$ per year (crypto trades every day; on the statistics of this annualisation see Lo, 2002). Testing many variants inflates the best Sharpe ratio by chance; the deflated Sharpe ratio (Bailey and López de Prado, 2014, 2012) is the probability that the true Sharpe exceeds the maximum expected under the null among N trials:

$$\mathrm{DSR} = \Phi\left(\frac{(\mathrm{SR} - \mathrm{SR}_0)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\,\mathrm{SR} + \frac{\hat{\gamma}_4 - 1}{4}\,\mathrm{SR}^2}} \right), \tag{13}$$

$$\mathrm{SR}_0 = \sqrt{V} \left[(1-\gamma)\,\Phi^{-1}\Big(1 - \frac{1}{N}\Big) + \gamma\,\Phi^{-1}\Big(1 - \frac{1}{Ne}\Big) \right],$$

where SR is the daily Sharpe ratio, T the number of daily returns, $\hat{\gamma}_3$ and $\hat{\gamma}_4$ their skewness and (non-excess) kurtosis, N the number of distinct trials registered for the strategy, V the variance of the daily Sharpe ratios across those trials, $\gamma \approx 0.5772$ the Euler–Mascheroni constant, e Euler's number and Φ the standard normal distribution function.

Pass criterion. Phase 1 (P1–P12) was registered at the start of the project with two conditions: at least 60% of positive months and a maximum drawdown below 10%; its deflated Sharpe ratio was computed and reported but was not yet part of the rule. From test A onwards the rule was at least 60% of positive months, maximum drawdown below 10% and DSR ≥ 0.95, and we use this stricter rule as the common yardstick in Table 1; no phase-1 variant met even the original two conditions (the best had 15% of positive months). For carry we added, before running it: funding collected minus costs > 0 in the real-funding period; no contract contributing more than 50% of the profit; the drawdown limit applied to both the real-funding period and the full period; and an ordered outcome rule (rejected if funding minus costs ≤ 0; rejected if the drawdown reached 10%; passed if all criteria held; otherwise inconclusive).

Pre-registration. Every test, its parameters and its decision rule were committed to the repository before it was run; after seeing results, only programming errors could be corrected (two were, in the Avellaneda–Lee engine, and both are documented), never parameters. All trials are stored and counted in N. The complete chronology is in Appendix A.

6 Results

6.1 The 16 registered tests

Table 1 reports all 16 tests and Figure 1 their capital curves. No test meets the criterion. Phase 1 (P1–P12) varied one parameter at a time around the base; test A (A1–A2) relaxed the multiple-testing correction and tried daily data with a one-year window; test B (B1) replaced pairs with Avellaneda–Lee residuals; and C1 is the funding-carry portfolio.

6.2 Why the pairs tests trade so little: the selection funnel

Figure 2 shows the selection funnel of the base configuration over the 221 cut-offs. Of 2,041 pairs tested within OPTICS clusters, 153 (7.5%) had $p < 0.05$ in both directions, against 102 expected if no pair were cointegrated; 41 survived Benjamini–Hochberg at 5%, 40 the spread filters and 13 the stability check in both halves of the window. Thirteen pairs were selected, in 12 of the 221 weeks. Relaxing the first filter multiplies candidates mostly with false positives (78 pairs with Benjamini–Hochberg at 10%; 282 with a single direction and no correction, against about 200 expected by chance). The bottleneck is cointegration itself.

The Kalman filter calibrated by maximum likelihood chose the smallest observation variance in the grid for all the selected pairs, $R = 0.01\,\sigma_{eq}^2$, with δ between 10^{-7} and 10^{-5}: the data prefer a filter that tracks the spread itself, so z^K measures one-hour surprises rather than departures from equilibrium; half of the base-case exits were such "breakdowns". Test A replaced it with a filter that updates only β and requires two consecutive bars, without changing the outcome.

6.3 Avellaneda–Lee

Method B finds candidates almost every week (204 of 221) instead of in 12, and trades 97 times. Its gross result is +42.5 against 36.0 of costs and +10.5 of funding: +16.9 summed over trades, and +1.8% (+17.7) at account level, the small difference being the part of the netted hedge book that is not attributed to individual trades. Exits at the signal earned +9.7 on average (59 trades), but 31 stops lost −17.0 on average because residuals of small contracts often drifted

Table 1: The 16 pre-registered tests (initial capital US$1,000). P1–A2: 25 July 2022 to 17 October 2026. B1 stopped at its emergency rule on 21 May 2024. C1: real-funding period (27 October 2025 to 17 October 2026); its full-period simulation is reported in Section 6.4. N: trials counted in the DSR.

ID	Description	Return	SR	DSR	N	Max DD	Pos. months	Trades
P1	Pairs, base	+0.1%	0.04	0.06	13	0.8%	6%	6
P2	volume $\geq$ US$0.5 M	+3.2%	0.79	0.50	13	1.6%	15%	19
P3	volume $\geq$ US$2 M	−0.6%	−0.38	0.00	13	1.4%	4%	8
P4	window 60 days	−0.05%	−0.02	0.05	13	0.8%	10%	13
P5	window 120 days	−1.9%	−0.82	0.00	13	2.1%	6%	7
P6	entry $\lvert z\rvert \geq 1.5$	+0.3%	0.14	0.09	13	1.1%	6%	9
P7	entry $\lvert z\rvert \geq 2.5$	−0.6%	−0.72	0.00	13	0.8%	2%	3
P8	4-hour bars	−1.4%	−0.76	0.00	13	1.6%	4%	7
P9	3 PCA factors	−0.5%	−0.36	0.01	13	0.8%	6%	6
P10	8 PCA factors	−0.1%	−0.06	0.04	13	0.9%	4%	7
P11	Kalman z for signals	−0.6%	−0.59	0.00	13	0.9%	6%	51
P12	estimated funding ×1.5	−0.9%	−0.58	0.00	13	1.4%	6%	7
A1	BH at 10%, volume $\geq$ US$0.5 M	+2.8%	0.63	0.34	14	2.2%	17%	24
A2	daily bars, window 365 days	−0.5%	−0.16	0.01	14	1.5%	2%	3
B1	Avellaneda–Lee residuals	+1.8%	0.09	0.06	15	10.4%	15%	97
C1	Funding carry (real-funding period)	+19.2%	1.32	0.90	1	9.3%	69%	98

SR: annualised Sharpe ratio. Positive months: share of the 52 calendar months for P1–B1 (for A2, 2% also of the 44 months from its first cut-off with a universe) and of the 13 months of the real-funding period for C1. For P1–P12 the stored DSR counted 13 trials because the code of the time also counted a duplicate trial run with identical parameters; with the later rule (one trial per parameter set) $N = 12$ and the best DSR changes from 0.503 to 0.501.

far more than their estimated equilibrium deviation, and the 10% emergency stop was hit on 21 May 2024 (Figure 5a).

6.4 Funding carry

Step 1 (persistence). In the real-funding period (50 weekly cut-offs, 27 October 2025 to 5 October 2026; 16 contracts per cut-off on median), the ranking of contracts by funding is clearly persistent: $\bar{\rho} = 0.436$ (Newey–West $t = 10.1$); a contract in the top tercile stays there the next week with probability 53%, and in the bottom tercile with probability 58% (Figure 3). The mean next-week spread was $D = 15.7\%$ a year ($t = 2.42$) against a break-even of 7.4%, so the registered rule to proceed was met. The mean is fragile, however: a single week (cut-off of 22 June 2026) contributed $D_c = 328\%$; without it the mean is 9.3%, and the median is 8.4%. Individual contracts revert quickly to their own mean (median daily AR(1) coefficient 0.37, half-life 0.7 days): what persists is the cross-sectional ordering, not the level.

Backtest. In the real-funding period the strategy returned +19.2% (annualised Sharpe 1.32, DSR 0.904 with $N = 1$), with a maximum drawdown of 9.3%, 69% positive months (9 of 13), and funding minus costs of +27.5 (funding +55.6, costs 28.2); the largest single-contract contribution was 33.5%. The full-period simulation, however, reached a drawdown of **10.07%** and its emergency stop fired on 30 May 2024. Under the outcome rule fixed in advance, the hypothesis is **rejected**. Two facts qualify this verdict and are discussed below: it is decided by 0.07 percentage points in the period with *estimated* funding, and with the real-funding period alone the outcome would have been "inconclusive" because the DSR (0.904) falls short of 0.95.

Three different quantities must not be confused here. The *observed return* of the real-funding period, +19.2%, is mostly price movement. The *premium attributable to the mechanism*—funding collected minus trading costs—was USD 27.5 on USD 1,000 of capital in 355 days, about 2.8% a year on capital, or 8.6% a year per unit of one-side notional (Section 7.4). The *statistical evidence* of an edge, a DSR of 0.904, did not reach the registered bar.

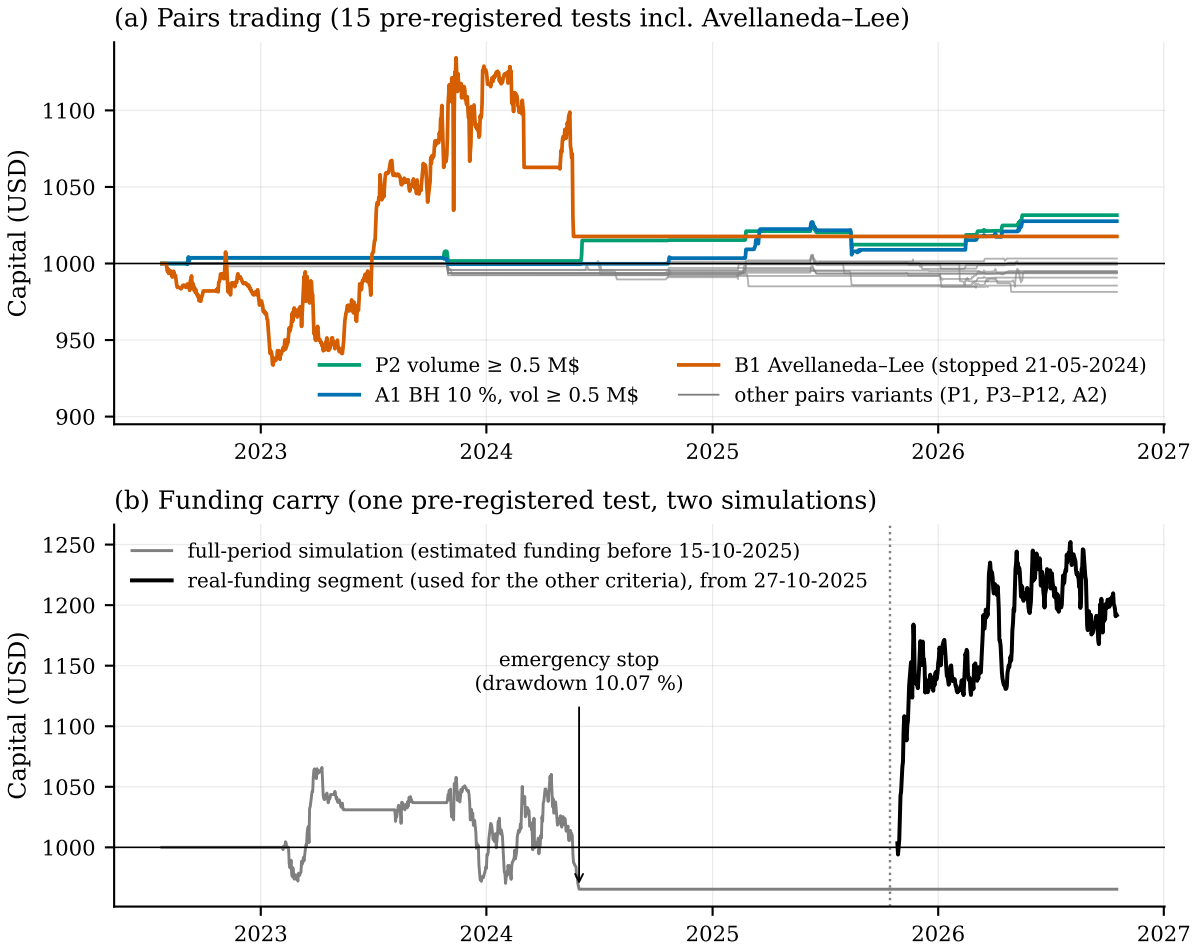


Figure 1: Capital curves of the pre-registered tests. (a) Pairs and Avellaneda–Lee. (b) Funding carry: the full-period simulation (grey; estimated funding before 15 October 2025) hits the 10% emergency stop in May 2024; the real-funding simulation (black) is the one that decides the other criteria.

Figure 5b decomposes the result: in the real-funding period, the profit came mostly from price moves of the legs (+207.7 on the legs and −43.5 on the BTC/ETH hedge), not from funding. During the 2023–2024 drawdown (Figure 4) the opposite happened: the short legs—contracts with the highest funding in a rising market—lost 193 on price while collecting 64 of funding, and 33 legs were closed by stop-loss, 24 of them shorts.

6.5 Reproducibility check

After reorganising the code to host several strategies, all fifteen earlier tests were re-run from the stored parameters and compared, without rounding, with the stored results (every metric, the capital of every day and every trade): all coincide exactly. Recomputing the weekly selections from scratch also reproduces them, with one documented exception: the base pairs selection had been computed while the data-collection process was writing new order-book snapshots; with the data as they stood before that cycle the recomputed selection is identical, while with the complete data the base variant would have returned −0.2% (10 trades) instead of +0.1% (6 trades). Neither version passes, but the episode illustrates how sensitive a low-frequency strategy is to small changes in estimated spreads.

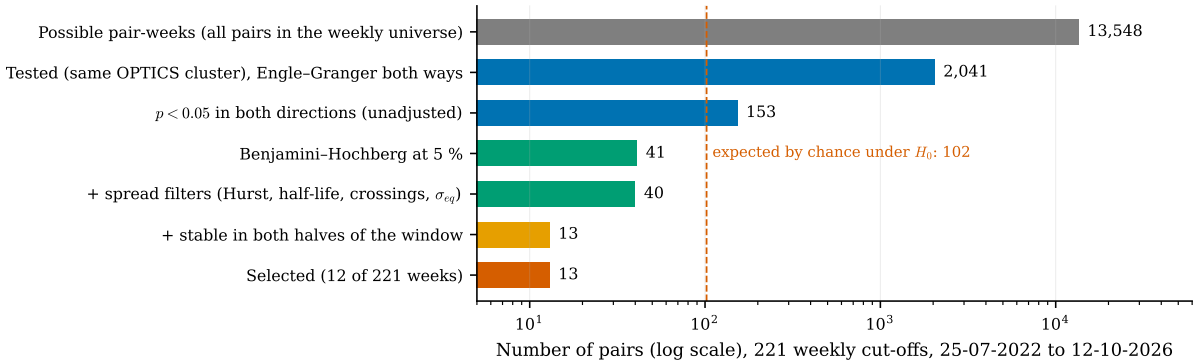


Figure 2: Selection funnel of pairs trading (base configuration), 25 July 2022 to 12 October 2026. The dashed line marks the number of pairs expected to show $p < 0.05$ in both directions if none were cointegrated.

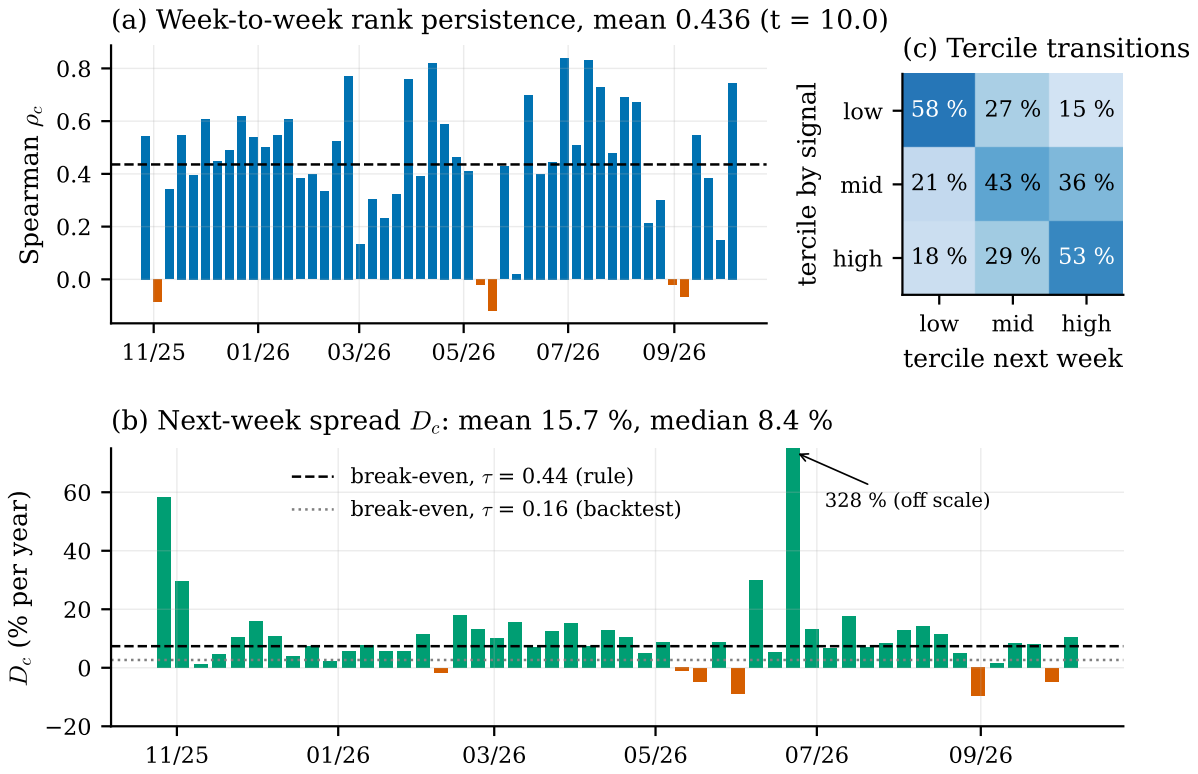


Figure 3: Persistence of funding in the real-funding period. (a) Spearman correlation between the ranking of contracts by their mean funding in one week and the next. (b) Next-week spread D_c between the three highest- and lowest-funding contracts, with the break-even spreads for the rotation of the registered rule ($\tau = 0.44$) and the one realised in the backtest ($\tau = 0.16$). (c) Tercile transition probabilities.

7 Four arguments

7.1 (a) The cost inequality

A trade with expected gross profit $\mathbb{E}[G]$ is worth taking only if

$$\mathbb{E}[G] \;>\; C \;=\; \sum_{k=1}^{K} c_{(k)}\,|N_{(k)}|, \tag{14}$$

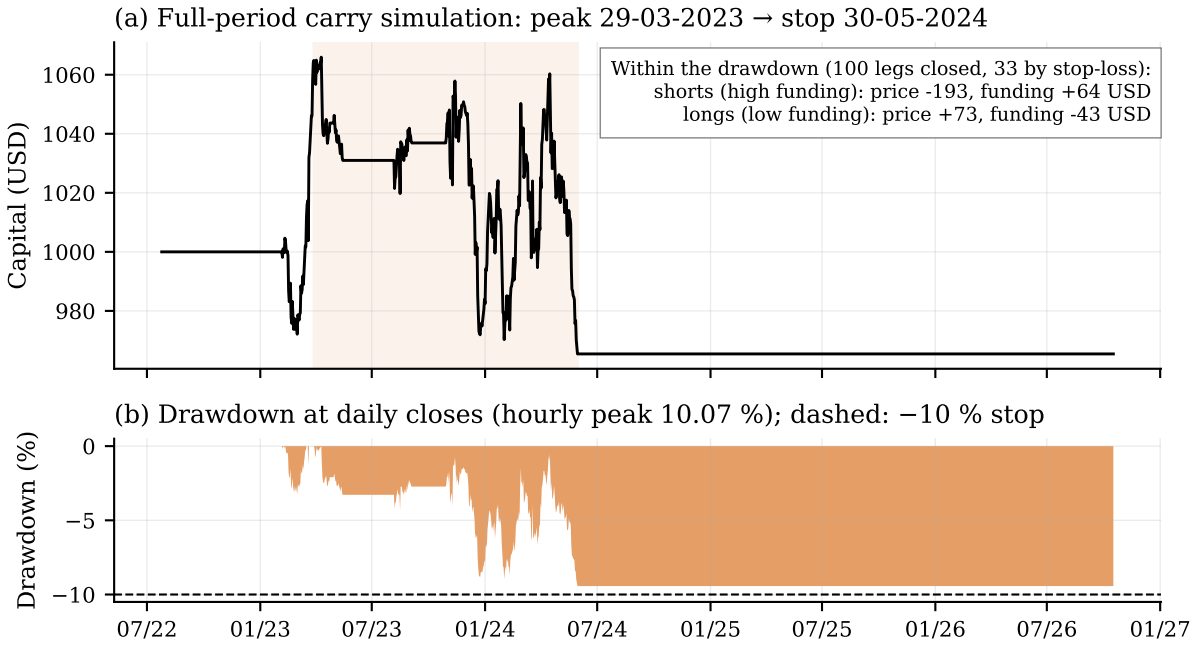


Figure 4: Full-period simulation of funding carry (estimated funding before 15 October 2025). The shaded area is the drawdown from the peak of 29 March 2023 to the emergency stop of 30 May 2024.

where K is the number of executions the trade requires, $N_{(k)}$ the notional of execution k and $c_{(k)}$ its cost per unit of notional, equation (4).

Avellaneda–Lee. Each trade needs $K = 6$ executions (opening and closing the asset and its BTC and ETH hedges). Per trade, with an asset leg of about US$100, we measured $\mathbb{E}[G] = 0.44$ against $C = 0.37$: costs absorb 85% of the gross edge. Moreover, the standard deviation of G across the 97 trades is 18.8, so the t-statistic of the mean of $G - C$ is 0.03—the edge left after costs is statistically indistinguishable from zero.

Carry. If n legs per side of notional N are held and a fraction τ of them is replaced every week (each replacement costs two executions), the annual cost is $52\,\tau\,(2n)\,2\,\bar{c}\,N$ and the annual funding income $n\,N\,F$, where $\bar{c}$ is the cost per execution and F the annual funding spread between the short and the long legs. The strategy covers its rotation costs only if

$$F \;>\; F^{*} = 208\,\tau\,\bar{c}. \tag{15}$$

With $\bar{c} = 0.08\%$, the tercile rotation $\tau = 0.443$ gives $F^{*} = 7.4\%$; the rotation realised by the banded portfolio, $\tau = 0.16$, gives $F^{*} = 2.7\%$. The measured median spread (8.4%) clears these thresholds, but not by a large margin.

Institutional contrast. The cost per execution, not the signal, is what a larger participant changes most. Fee schedules fall with trading volume (Kraken, 2020; Binance, 2026), and a liquidity provider that executes passively earns rather than pays the half-spread (Harris, 2003). As a purely illustrative scenario (not a quote of any institution's costs), if $\bar{c}$ fell from 0.08% to 0.03% (a maker fee of 0.02%, the base-tier rate on Binance (Binance, 2026), plus slippage), the Avellaneda–Lee cost per trade would fall from 0.37 to about 0.14 and F^{*} by a factor of 2.7. The costs of Do and Faff (2012) for U.S. pairs trading illustrate the same point: profitability depends on who pays which costs.

7.2 (b) Breadth: the fundamental law of active management

Grinold (1989) showed that the information ratio of an active strategy is approximately

$$\mathrm{IR} \approx \mathrm{TC} \cdot \mathrm{IC} \cdot \sqrt{\mathrm{BR}}, \tag{16}$$

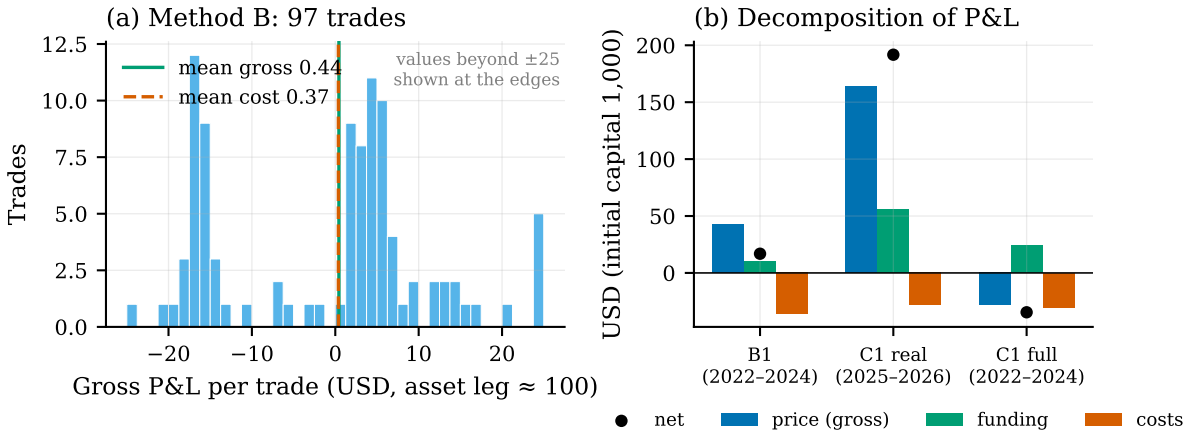


Figure 5: (a) Gross profit per trade of the Avellaneda–Lee strategy (97 trades): the mean gross gain barely exceeds the mean cost. (b) Decomposition of profit and loss into price, funding and costs for B1 and for the two carry simulations (US$, initial capital 1,000).

where IC (information coefficient) is the correlation between forecasts and subsequent residual returns, BR (breadth) the number of independent bets per year, and TC ≤ 1 the transfer coefficient that measures how much of the forecast survives portfolio constraints (Grinold and Kahn, 2000; Clarke et al., 2002). To reach IR $= 1$ with TC $= 1$, a strategy needs $\mathrm{IC}^* = 1/\sqrt{\mathrm{BR}}$.

Our base pairs strategy traded 1.4 times a year ($\mathrm{IC}^* \approx 0.85$, an impossible forecasting skill); Avellaneda–Lee traded 53 times a year ($\mathrm{IC}^* \approx 0.14$). For carry, with about 16 eligible contracts ranked every week, BR $\approx 16 \times 52 = 834$ and $\mathrm{IC}^* \approx 0.035$. We estimated, after the fact and only descriptively, the weekly rank correlation between the carry signal and the realised one-week result of being short each contract (funding received minus residual price change): IC $= 0.042$ with a standard error of 0.038. By equation (16) this corresponds to IR ≈ 1.2, but the two-standard-error interval spans -1.0 to 3.4. Decomposed, the signal predicts next-week *funding* very well (IC $= 0.43$, $t = 10.2$) and next-week *price* not at all (IC $= 0.02$, $t = 0.56$).

Institutional contrast. Avellaneda and Lee (2010) traded a broad universe of U.S. equities daily; a universe such as the Russell 1000 contains 1,000 large companies (FTSE Russell, 2026), so 1,000 names rebalanced on 252 trading days give BR in the order of 2.5×10^5 and $\mathrm{IC}^* \approx 0.002$ (Figure 6). Our universe after liquidity filters averaged 9.9 contracts. Bets in crypto are also strongly correlated through a common market factor (Liu et al., 2022), so the effective breadth is smaller still. A larger participant can widen the universe—more venues, spot markets, lower liquidity thresholds that its execution can tolerate—in ways a retail account on one venue cannot.

7.3 (c) Statistical power: how good must a strategy be to be recognised?

Setting DSR $= 0.95$ in equation (13) with normal returns ($\hat{\gamma}_3 = 0$, $\hat{\gamma}_4 = 3$, for which the denominator is close to 1) gives the minimum annualised Sharpe ratio that a track record of T days must show:

$$\mathrm{SR}^*_{\mathrm{annual}} \approx \sqrt{365}\left(\mathrm{SR}_0 + \frac{z_{0.95}}{\sqrt{T-1}}\right), \tag{17}$$

with $z_{0.95} = 1.645$ the 95th percentile of the standard normal and SR_0 as in equation (13). With a single trial ($\mathrm{SR}_0 = 0$) the threshold is 1.65 for twelve months and 3.33 for three months (Table 2, Figure 7). Carry's real-funding Sharpe of 1.32 over 355 days falls short of the 1.67 required, which is why its DSR is 0.904. Conversely, if returns were normal and the true Sharpe were exactly 1, the probability that a twelve-month record clears the bar would be only 26% (12% with three months); with four years of data, 66%. Testing many variants raises SR_0:

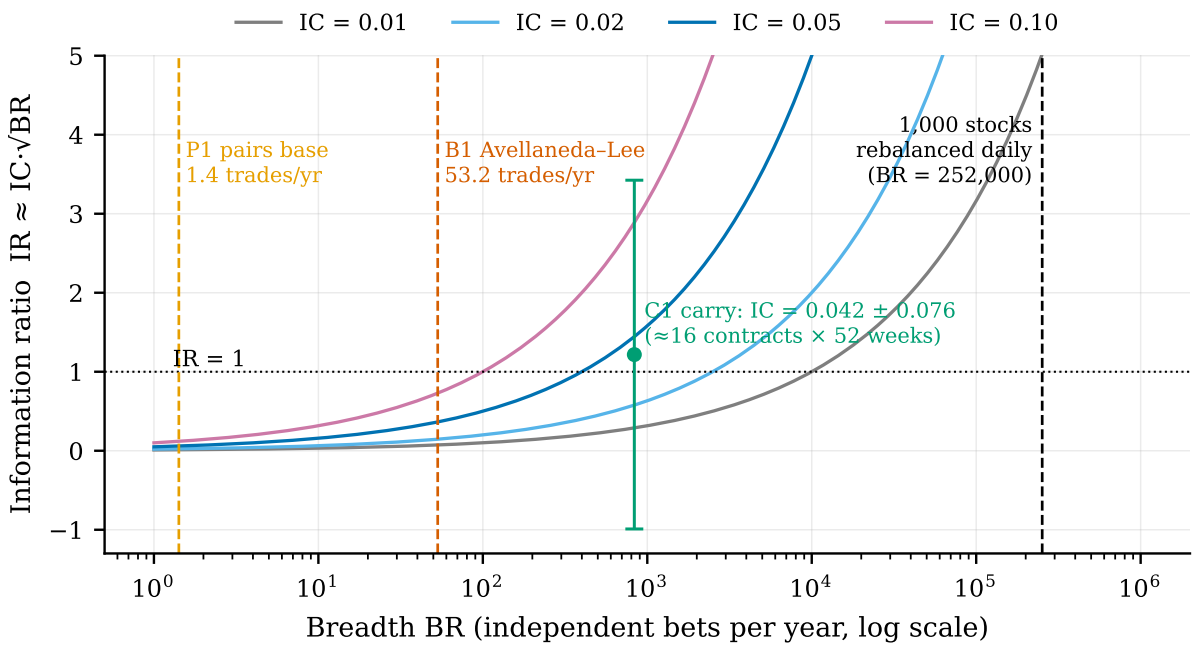


Figure 6: The fundamental law: information ratio as a function of breadth for several information coefficients. Our strategies sit at very low breadth; the carry point shows the post-hoc estimate of its IC with a two-standard-error band.

with the dispersion of our fifteen pairs trials, 16 trials raise the twelve-month threshold from 1.65 to 2.52.

Table 2: Minimum annualised Sharpe ratio for DSR ≥ 0.95, equation (17), as a function of track-record length T and number of trials N (V = variance of the daily Sharpe ratios of our 15 pairs trials).

T (days)	90	182	365	730	1,546	3,650
$N = 1$	3.33	2.34	1.65	1.16	0.80	0.52
$N = 16$	4.20	3.21	2.52	2.04	1.67	1.39

Institutional contrast. The constraint is not specific to retail; it is about data. Avellaneda and Lee (2010) report eleven years of daily backtests, and established managers accumulate long live records and can test new ideas on decades of data in deep markets (on multiple testing in finance, see Harvey et al., 2016; Bailey et al., 2014). Our strategies depend on funding data that exist only for twelve months, and on a market a few years old; a three-month live simulation, however useful to validate the machinery, cannot statistically confirm profitability.

7.4 (d) Premium versus risk in funding carry

Consider a carry book with n legs per side, each of notional N. Let π be the annual net premium (funding collected minus costs) per unit of one-side notional, and let the beta-hedged price return of each leg have annual volatility σ and an average pairwise correlation $\bar{\rho}$ with the others. Writing each leg's return as a common component plus an idiosyncratic one, the common component cancels between the long and the short side, the variance of the book's price result is $2n\,N^2\sigma^2(1-\bar{\rho})$ and its expected premium is $nN\pi$, so the Sharpe ratio of the premium is

$$\mathrm{SR}_n = \frac{\pi\sqrt{n}}{\sigma\sqrt{2\,(1-\bar{\rho})}}. \tag{18}$$

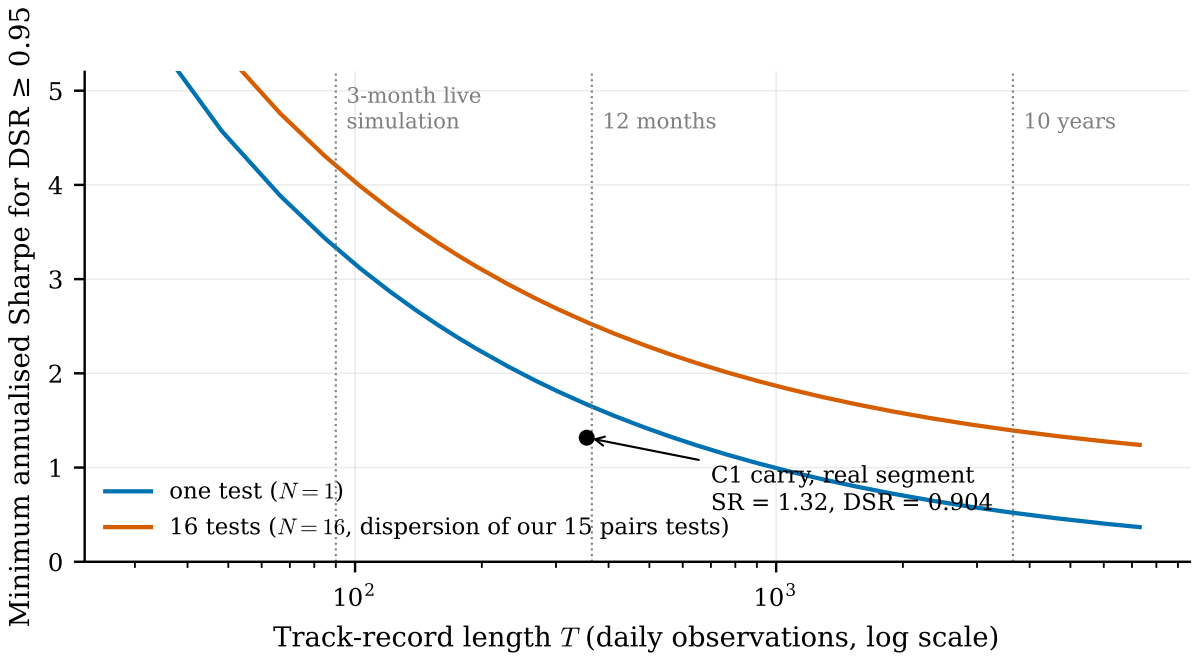


Figure 7: Minimum annualised Sharpe ratio required for a deflated Sharpe ratio of 0.95, as a function of the length of the record, for one trial and for sixteen trials.

In the real-funding period we measured $\pi = 8.6\%$ per unit of one-side notional (US$27.5 of funding minus costs on roughly US$330 per side), $\sigma = 59\%$ and $\bar{\rho} = 0.21$ (the last two descriptively, from weekly beta-hedged returns of the 17 contracts with at least 20 weeks in the universe). With our $n = 3$, $\mathrm{SR}_3 = 0.20$. Relative to capital, the premium was 2.8% a year, while the volatility of the strategy was 15% a year and a 10% drawdown would take about 3.5 years of premium to recover. The realised Sharpe of 1.32 is therefore mostly the result of favourable price moves, consistent with the price IC of 0.02 above, and it reversed in the 2023–2024 rising market (Figure 4).

Institutional contrast. Equation (18) shows the two levers that capital provides. Breadth: with the same premium and risk, SR_n reaches 0.81 with $n = 50$. In our study $n = 3$ was a registered design choice, but the binding limit is the market rather than capital: fifty legs per side need at least a hundred eligible contracts, against about sixteen in our carry universe, and with US$1,000 at a gross leverage of at most 2 they would also mean legs of about US$20, a size at which we have not checked minimum order sizes. Cost: under the illustrative 0.03% per execution, π rises to about 14% and SR_n reaches 1.65 with $n \approx 77$ (Figure 8). Institutions can also implement carry as cash-and-carry against spot, where the price risk largely cancels and in-sample Sharpe ratios of 7–10 have been reported (Christin et al., 2022); the size of that premium, however, varies strongly over time, is linked to retail demand for leverage and scarce arbitrage capital, and comes with crash risk (Schmeling et al., 2023; Brunnermeier et al., 2009), and it has been compressing since 2024 (Borri et al., 2025). The cross-section of perpetual returns has been linked to the futures–spot basis (Cao et al., 2024), which supports looking at funding, but our evidence is that the premium alone was too small for the risk carried by a three-legs-per-side book.

7.5 Synthesis: how the four arguments combine

The four arguments can be read as separate obstacles, but they act on the same quantity: the evidence that a strategy accumulates per bet. Think of a strategy as a sequence of bets. Let m be the expected net result of one bet, s_b its standard deviation and $\varsigma = m/s_b$ its signal-to-noise

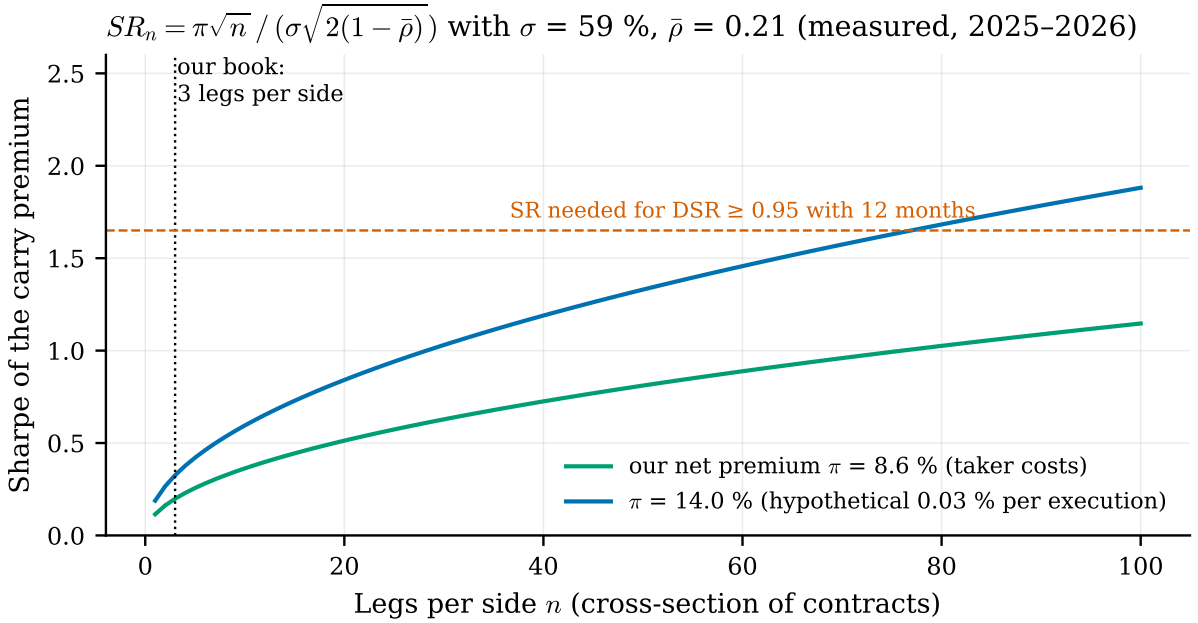


Figure 8: Sharpe ratio of the carry premium, equation (18), as a function of the number of legs per side, with the measured premium and with a hypothetical lower cost per execution. The dashed line is the Sharpe ratio required for a DSR of 0.95 with twelve months of data and one trial.

Table 3: Summary of the four arguments: retail conditions measured in this study versus institutional conditions (sources in the text; the 0.03% cost is an illustrative scenario).

Argument	This study (retail, Kraken, 2022–2026)	Institutional participant
(a) Costs	0.08% per execution (taker); B1: gross 0.44 vs cost 0.37 per trade, $t(G-C) = 0.03$; carry break-even 2.7–7.4%	Volume-tiered fees, passive execution; e.g. 0.03% per execution cuts B1's cost to 0.14
(b) Breadth	9.9 contracts on average; 1.4 (P1), 53 (B1) and ≈834 (C1) bets a year; required IC 0.85 / 0.14 / 0.035	Broad universes (e.g. 1,000 names daily, BR $\sim 2.5 \times 10^5$, required IC ≈ 0.002)
(c) Power	12 months of real funding: SR ≥ 1.65 needed with one trial (3.33 with 3 months); carry's 1.32 gives DSR 0.904	Long histories and live records lower the bar
(d) Premium vs risk	$\pi = 8.6\%$, $\sigma = 59\%$, $\bar{\rho} = 0.21$, $n = 3$: $SR_3 = 0.20$; 2.8% a year on capital vs 15% volatility	More legs and lower costs raise SR_n; cash-and-carry against spot

ratio (a "Sharpe ratio per bet"). After n independent bets, the t-statistic of the average result, and the number of bets needed for the edge to stand out from noise, are

$$t = \varsigma\sqrt{n}, \qquad n^* = \left(\frac{z}{\varsigma}\right)^2, \qquad T^* = \frac{n^*}{\mathrm{BR}}, \tag{19}$$

where z is the critical value of the test ($z = 1.645$ for 95% confidence, one-sided, before any correction for multiple testing) and T^* the years needed by a strategy that makes BR independent bets a year. Equation (19) is the fundamental law of active management, equation (16), seen from the other side: Grinold's law says how fast an edge pays; equation (19) says how long it takes to establish that it exists. Two assumptions matter. The bets must be independent; correlated assets or overlapping trades provide fewer effective bets than their number. And n^*

uses the measured edge as if it were the true one: it answers "how much evidence would this edge need", not "is the edge real".

Table 4 applies equation (19) to our own trades. Avellaneda–Lee's measured edge, 0.17 per trade after costs and funding against a dispersion of 19, would need about 33,000 independent trades to stand out from noise: more than six centuries at the pace of our account, and about seven weeks for a participant making 252,000 bets a year (1,000 names rebalanced daily). The best pairs variant, P2, had a far larger edge per trade ($\varsigma \approx 0.32$) but traded 19 times in four years; its raw evidence ($t \approx 1.4$) is about what the best of thirteen variants would show by chance, which is what its DSR of 0.50 says. In carry the measured edge per leg was larger again, but its legs are held together each week and are not independent, so the 320 legs of the table understate the evidence needed.

Table 4: Bets needed for an edge to stand out from noise, equation (19) with $z = 1.645$, taking each strategy's measured net result per trade (USD, asset leg ≈ 100) as if it were the true one. Bets per year: our active trading pace. Time needed: at that pace, and for a participant making 252,000 bets a year.

							Time needed	
Test	Trades	Mean	Std. dev.	ς	n^*	Bets/yr	At our pace	At 252,000/yr
P1 pairs, base	6	0.10	4.9	0.020	6,600	1.4	≈4,700 years	10 days
B1 Avellaneda–Lee	97	0.17	19.2	0.009	33,000	53	≈620 years	7 weeks
C1 carry (legs)	98	2.49	27.0	0.092	320	101	≈3 years	11 hours

Carry legs are not independent, so its n^* is optimistic; Section 7.2 counts about 834 ranking decisions a year for carry, whereas here we count the legs actually opened. P1's six trades give a very imprecise estimate of ς; the row is illustrative.

Costs act on the square. Costs are subtracted from the edge per bet before it is squared. If the gross edge per bet is g and the cost per bet is C, ignoring funding, the evidence needed grows by the factor

$$\frac{n^*_{\text{net}}}{n^*_{\text{gross}}} = \Big(\frac{g}{g - C}\Big)^2. \tag{20}$$

For Avellaneda–Lee, $g = 0.44$ and $C = 0.37$ (0.438 and 0.371 before rounding): costs that take 85% of the edge multiply the evidence needed by about 44; under the illustrative cost of 0.03% per execution of Section 7.1 ($C \approx 0.14$) the factor would be about 2. A cost is therefore not only a fee on profits but also a tax on the speed at which an edge can be established, which is why cost reductions matter most to frequent, small-edge strategies.

The stop can arrive before the proof. For a strategy whose expected return is small relative to its volatility σ, the expected maximum drawdown after T years is close to that of a Brownian motion without drift (Magdon-Ismail et al., 2004),

$$\mathbb{E}\left[\text{MDD}_T\right] \approx \sqrt{\pi/2}\,\sigma\sqrt{T} \approx 1.25\,\sigma\sqrt{T}, \tag{21}$$

where MDD_T is the largest fall from a previous peak within T years. With the 15% annual volatility of our carry portfolio, the expected drawdown reaches the 10% limit after $T \approx \big(0.10/(1.25 \times 0.15)\big)^2 \approx 0.28$ years, about three and a half months, whereas the evidence of Table 4 needs years. A prudent drawdown limit combined with a low ratio of premium to volatility makes it likely that a strategy is stopped before its edge, if any, can be demonstrated. Diversification lowers σ, and the time to an expected stop grows as $1/\sigma^2$.

What scale changes, and what it does not. Table 5 lists the variables that enter these equations and the kind of constraint that sets each of them. Capital raises the number of legs and lowers the cost per bet; it does not lengthen the history of funding data, add liquid contracts or remove their common movement. Institutions are not exempt either: their costs are lower but not zero, large orders move prices, and the small contracts that offered the

highest funding in our sample could absorb only small positions, so capacity limits what a large participant could earn from the same signals. The synthesis is therefore not a proof that an individual cannot profit from statistical arbitrage; it shows how, in our case, implementation constraints absorbed part of each edge and kept the rest below the threshold at which it could be distinguished from luck.

Table 5: The variables that decide the arithmetic of an edge, the kind of constraint that sets them, our values and illustrative institutional orders of magnitude (Section 7).

Variable	Where it enters	Set by	This study (retail)	Institutional participant
Independent bets per year	$t = \varsigma\sqrt{\mathrm{BR}\,T}$	Market (liquid contracts, common factor), capital, design	1.4 (P1), 53 (B1), ≈100 correlated legs (C1); about ten liquid contracts	Hundreds of thousands, pooled across markets and strategies
Years of evidence	$t = \varsigma\sqrt{\mathrm{BR}\,T}$	Data	Twelve months of real funding	Decades of data; long live records
Cost per bet	$(g/(g-C))^2$	Capital (volume tiers), design (taker)	0.08% per execution, always paying the spread	Lower tiers; passive execution can earn the spread
Legs per side	$\mathrm{SR}_n \propto \sqrt{n}$, eq. (18)	Design, market (eligible contracts), capital (leverage cap)	Three per side (registered choice); about 16 eligible contracts	Tens or hundreds, within capacity
Drawdown limit	eq. (21)	Design (risk policy)	10% emergency stop	Risk budgets across many strategies

8 Discussion

Pairs trading failed mainly because cointegration among the few liquid contracts is barely distinguishable from chance (153 rejections against 102 expected), leaving one or two trades a year: too few to cover costs reliably and far too few for any statistical test. The universe is the deeper cause: with about ten contracts, there are about 61 possible pairs a week, and a market dominated by a common factor produces few stable relationships. The base variant's dependence on estimated spreads (Section 6) shows how fragile such low-frequency results are.

Avellaneda–Lee solves the breadth problem partially—candidates every week—but not the cost problem: six executions per trade at retail cost consume most of a gross edge that is small relative to the dispersion of outcomes, and heavy-tailed residuals of small contracts produce large stop-losses.

Funding carry is the most informative case. The economic premise holds: funding is persistent across contracts, and the funding collected covered the costs. But the premium is small relative to the price risk of a few legs, and its sign is not protected against market regimes: in a rising market, the contracts with the highest funding are those with the most leveraged long demand, and shorting them was painful. This is the crypto analogue of the crash risk of currency carry (Brunnermeier et al., 2009). The verdict sits at the margin and was decided in the period with estimated funding; we report it as a rejection because that was the rule we fixed in advance, and we interpret it as "no evidence of a premium large enough for the risk at this scale", not as evidence that funding carry cannot work.

Taken together, the four arguments are multiplicative rather than independent: high costs lower the net edge, the small universe limits breadth, short histories make any remaining edge statistically invisible, and a small universe and small capital limit the diversification that would raise the Sharpe ratio of a modest premium. A large institution relaxes several of them

at once—lower costs, broader universes across markets and asset classes, longer records pooled across strategies, more legs—but not all, and not for free: the short history of funding data and the small number of liquid contracts are properties of this market, and large participants face costs, execution problems and capacity limits of their own. Section 7.5 shows the common arithmetic behind the four arguments and which constraints capital can and cannot remove.

9 Limitations

Pending estimates. The comparison with European shares in Section 3 uses verified commission schedules but does not include borrowing fees, exchange and clearing fees, market-data subscriptions or the cost of survivorship-free historical data, so it is a lower bound; the platform comparison of 23 September 2026 relies on secondary sources; and we have not confirmed in an official document that Kraken's published futures fee table applies unchanged to its European entity.

- **Only twelve months of real funding.** Before 15 October 2025 funding is estimated; the estimate orders contracts well but its level has a mean absolute error of 11.4% a year, and selecting the highest estimated funding favours positive estimation errors (a winner's curse). The carry verdict was decided in this estimated period.
- **The carry verdict is at the margin** (10.07% against a 10% limit). With the real-funding period alone the outcome would have been inconclusive, not a rejection.
- **The mean funding spread of step 1 depends on one week** (328% annualised); without it the mean is 9.3%, still above the registered break-even of 7.4%, but close.
- **Small universe and a single venue**: about ten liquid contracts; results may differ on venues or periods with broader liquid cross-sections.
- **Sensitivity to estimated spreads**: the base pairs variant changes from +0.1% to −0.2% with a small change in the order-book snapshots used to estimate spreads; the order-book history is short and past weeks use spreads measured in October 2026.
- **Execution model**: taker-only fills at the hourly close, a fixed slippage, hourly stop checks (realised stops often exceeded the 1.5% threshold within an hour) and no explicit market impact. The data snapshot ends on 17 October 2026; for the last three hours some prices were missing in the cached matrices, with negligible effect.
- **Post-hoc measures**: the information coefficients and the volatility and correlation used in Section 7 were not pre-registered; they are descriptive and were not used for any decision.
- **Period**: 2022–2026 includes a bear market, a recovery and the period of compressed carry premia documented by Borri et al. (2025); other periods may behave differently.

10 Future work: pre-registered hypotheses

Each hypothesis below will be registered in the repository (with its parameters, data and pass rule) before any data are examined, and evaluated only with data generated *after* the publication of this paper, so that no observation used here can inform it. Live validation will follow the rule already fixed for carry: a simulation without money whose goal is to validate the system (the live portfolio equals the backtest's each week; simulated funding equals actual funding; measured costs at most 25% above the model; no drawdown of 10%; funding minus costs positive), not to prove profitability.

H1: regime filter for carry. A two-state Markov-switching model (Hamilton, 1989) fitted to weekly market returns and the cross-sectional dispersion of funding classifies each week as "calm" or "stressed"; carry is active only when the filtered probability of the calm state exceeds 0.5 at the cut-off. Prediction: a lower maximum drawdown than unconditional carry with a similar funding premium.

H2: passive execution. Limit orders placed inside the spread reduce the cost per execution below 0.04% with a fill rate above 70%, measured in live simulation with the real order book.

H3: breadth within constraints. With capital that allows five legs per side above minimum order sizes, the Sharpe ratio of the carry premium increases as predicted by equation (18).

11 Conclusion

On Kraken's linear crypto perpetuals, with retail costs and capital and 2022–2026 data, none of 16 pre-registered statistical-arbitrage tests—cointegration-based pairs, Avellaneda–Lee residuals and cross-sectional funding carry—met a pass criterion fixed in advance. Four quantitative arguments explain the outcome: costs absorb most of the gross edge, the liquid universe provides very little breadth, twelve months of data cannot distinguish a moderate edge from luck, and the funding premium is small relative to the price risk of a few legs. Several of these constraints are relaxed by the scale, costs and diversification of institutional participants; others belong to the market and its data. These conclusions apply to these strategies, this venue, retail costs and this period; they do not establish that statistical arbitrage is impossible in general.

To the question that motivates this paper, our answer is specific rather than universal. Using the same models as a quantitative fund did not reproduce a fund's results because, in our implementation, the conditions around the models changed what they could deliver: costs per execution absorbed most of the gross edge of the residual strategy; a small, correlated universe provided few independent bets; twelve months of real funding data could not distinguish a moderate edge from luck; and a funding premium of about 2.8% a year on capital was small next to the price risk of three legs per side, under a drawdown limit that, at that volatility, is expected to be reached long before such evidence can accumulate. Some of these constraints depend on capital, others on the market and its data, and others on our own design; more capital alone would not remove all of them, and institutions face limits of their own. Not passing our tests does not show that the true return of these strategies is zero.

Use of AI tools and authorship

The trading-research system (data collection, backtesting engines, tests, reproduction checks) and drafts of this paper were developed with Claude Code (Anthropic, 2026), an AI coding assistant by Anthropic, working under the author's direction. The research question, the hypotheses, the parameters, the risk limits, the pre-registration rules and every decision about what to test and how to judge it were made by the author, who reviewed the results and the text. Errors remain the author's responsibility.

Disclaimer

This paper is for research and educational purposes only. It is not financial, investment, legal or tax advice, and nothing in it is a recommendation to buy or sell any financial instrument.

Derivatives on crypto-assets are complex and high-risk; past or simulated performance is not indicative of future results.

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A Reproducibility

Code and results. The repository contains the full system (Python 3.14): data agents, cleaning, universe construction, the strategies (`estrategias/pares`, `estrategias/carry`), a generic walk-forward engine (`backtest/`), 124 automated tests, and the scripts that produce every number and figure of this paper from the stored results (`publicacion/scripts/`). Raw market data from Kraken are not redistributed; they can be downloaded from the public API with the data agent.

Chronology of the registered tests.

- Phase 1 (P1–P12), 17–18 October 2026; executions `20261018-004946-f40d` (a trial run of P1) and `20261018-043054-b7cb`.
- Test A (A1–A2), registered in commit `a418894` before execution `20261018-104521-fa1d`.
- Test B (B1), registered in commit `55debfd`; two programming errors corrected (commits `7c7c8d9` and `14b8078`); final execution `20261018-110331-8974`. All three executions share the same parameters and count as one trial.
- The state of the project at the end of the pairs research is tagged `v1.0-pares-fase1`.
- Carry: design approved and step 1 registered in commit `78e7ba6`; backtest rules registered in `9c4a1c4`; code in `a6dfe2f`; execution `20261019-084717-a94d`.

Reproducing the results. Each stored execution can be re-run and compared with the stored results without saving anything, for example

```
python -m backtest.ejecutar -estrategia pares
       -reproducir 20261018-043054-b7cb -referencia 2026-10-17T22:51:10Z
```

where the reference date fixes the cut-offs, the order-book measurements and the exchange rate as they were when the test was run. The aggregated results and figures of this paper are produced by `python -m publicacion.scripts.extraer_resultados` (read-only) and `python -m publicacion.scripts.figuras`.